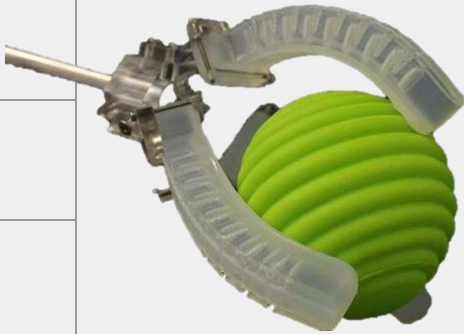


JAMMING GRIPPER FOR HARVESTING

By: Therese Joseph
Supervisors: Frederic Maire and David Howard



SOFT ROBOTICS



FLEXIBLE/COMPLIANT

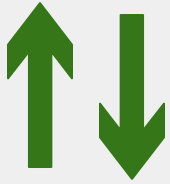


ROBUST TO IMPACT



**SIMPLE CONTROL
MECHANISM**

JAMMING



Transition from fluid like – solid like state



Achieved through change in confining volume



GAPS WITHIN THE FIELD



Potential for jamming hasn't been explored in automated harvesting

Most designs for robotic grippers in agriculture utilise a hand design with 2-3 fingers

Jamming can be particularly useful as the viscoelasticity of fruits changes based on species, ripeness and type of fruit

DESIGN GOALS

OPERATION RANGE

Applicable to crops with variations in size, shape and texture, targeted towards high value crops

EASE OF CONTROL

Utilising the optimal number of actuators with simple controllers

MINIMUM DAMAGE

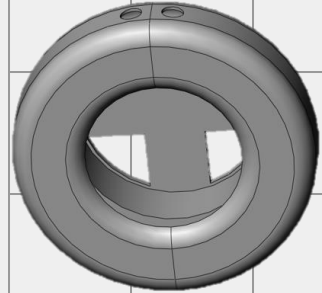
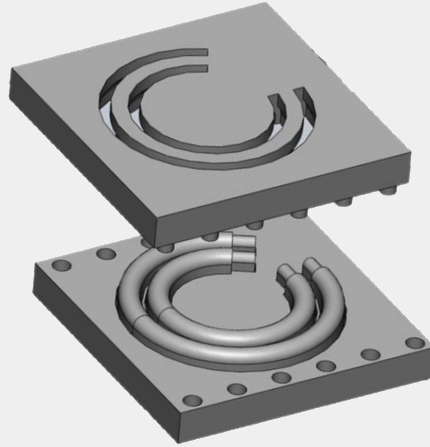
Enable soft gripping to avoid bruises and cuts



MANUFACTURING



3D PRINTING



SILICONE CASTING



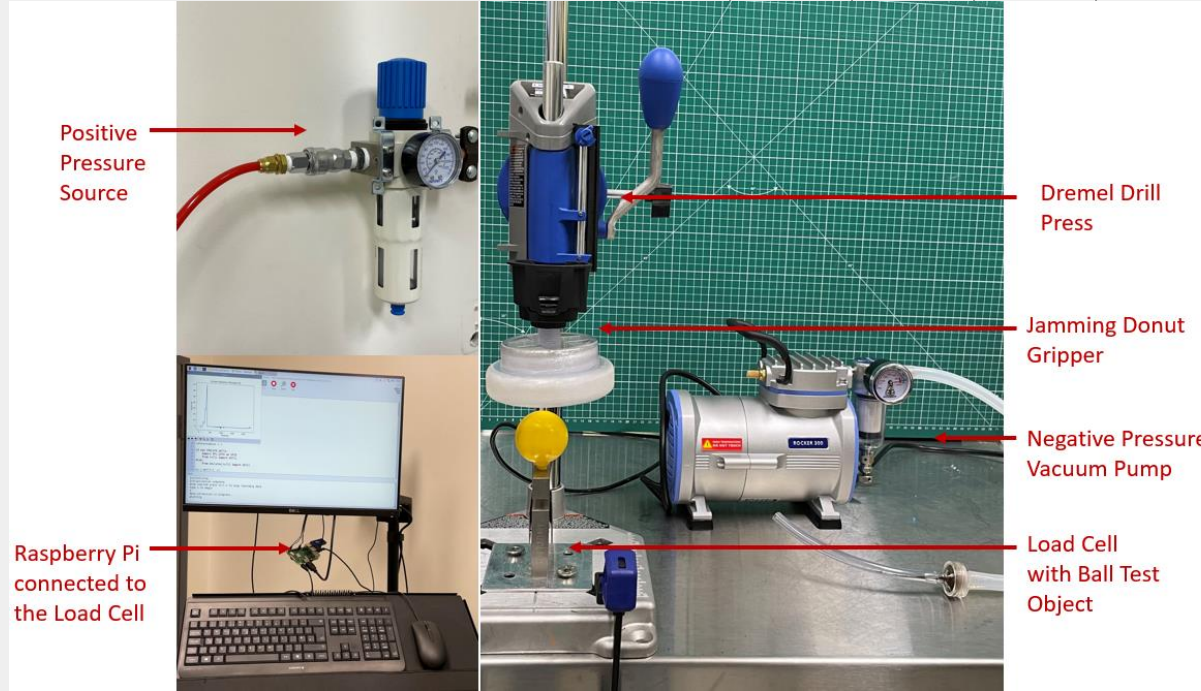
FINAL PRODUCT

Jamming Donut Gripper



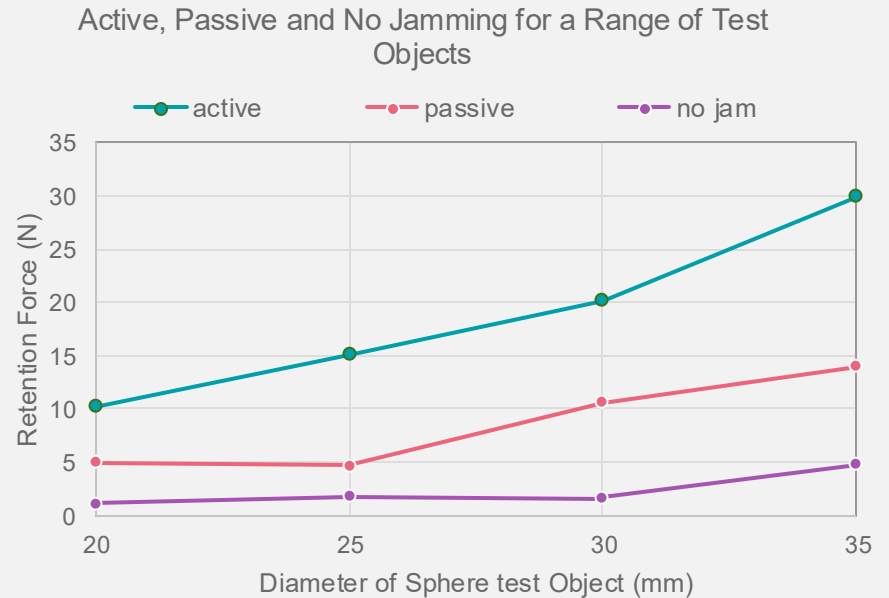
TEST SETUP

Test Objects



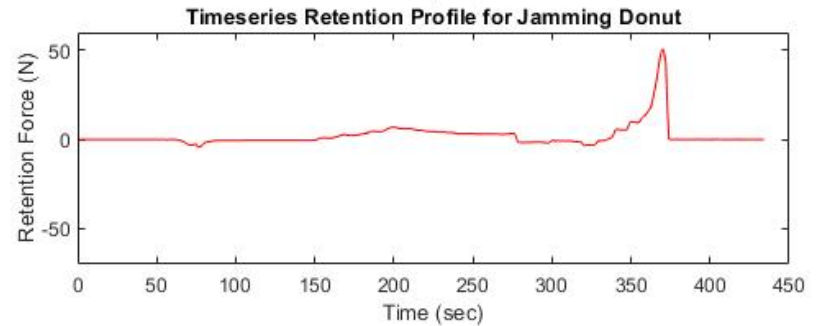
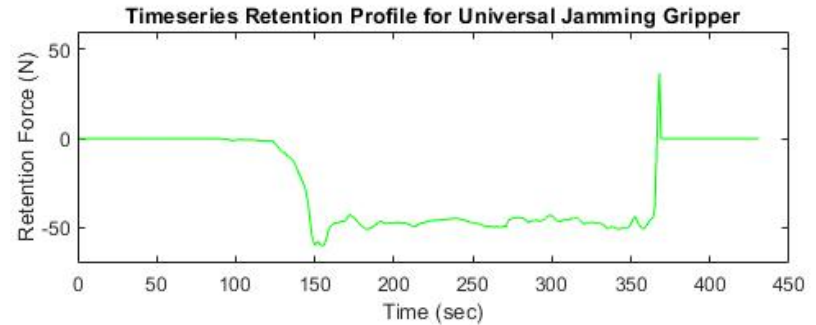
RESULTS

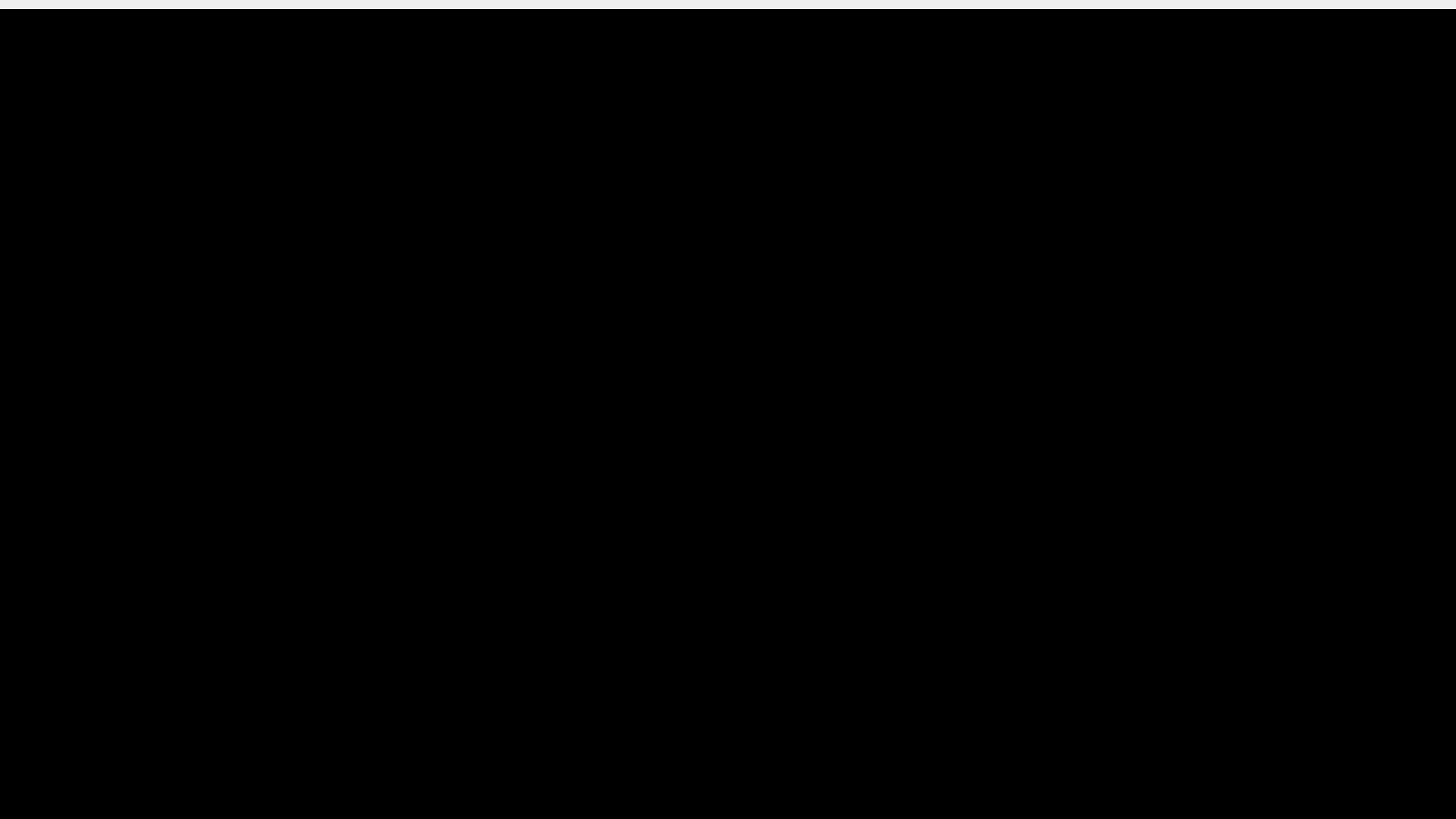
1. Conducted retention force test with objects ranging from 20mm-35mm
2. Repeated the tests for three states
 - No jamming
 - Passive Jamming
 - Active jamming



RESULTS

1. Comparison of traditional jamming gripper versus free space jamming.





MILESTONES

LITERATURE REVIEW

1

Agriculture
robotics and
soft robotics

PROTOTYPE DESIGN

2

Torus structures,
winding coil,
pneumatic actuator

FABRICATION

3

Cast molding
and shell
construction

EXPERIMENT/ DATA COLLECTION

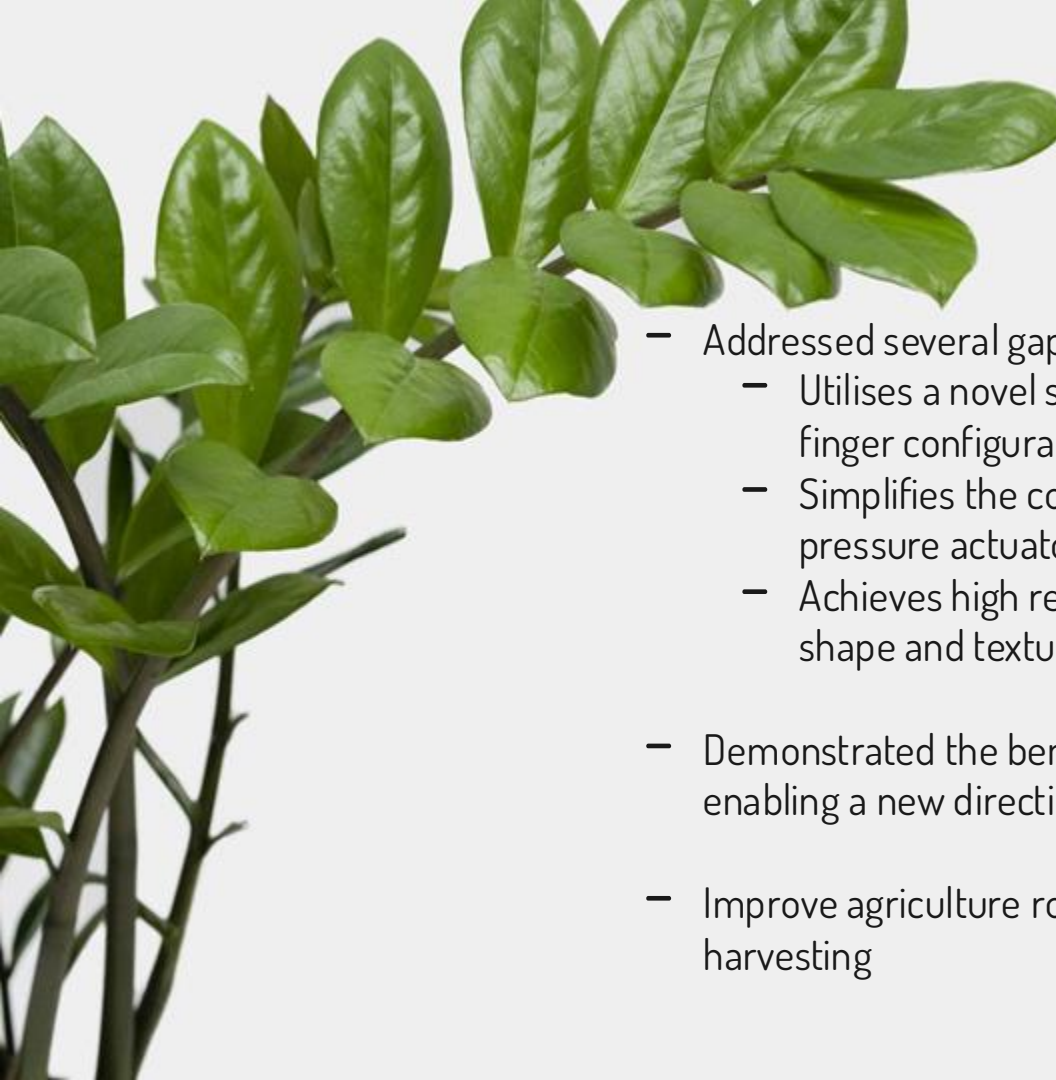
4

Retention
force test with
test objects

ASSESSMENTS

5

Proposal, update,
presentation,
report



SUMMARY

- Addressed several gaps within harvesting robotics
 - Utilises a novel structure straying away from typical 2-3 finger configuration
 - Simplifies the control architecture by only requiring two pressure actuators
 - Achieves high retention for crops with variation in size, shape and texture.
- Demonstrated the benefits of free space jamming for harvesting, enabling a new direction for research within soft robotics
- Improve agriculture robotics by enabling consistent and efficient harvesting